Answering questions spontaneously

Practical guide for subject lecturers | D06-S1–S6

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Answering a question in class involves understanding the question, retrieving subject knowledge and putting an answer into words. Students may need support with one or more of these steps.

Use the approaches below to help students demonstrate their understanding. A hesitant response does not, by itself, show a lack of subject knowledge.

1. Ask one clear question at a time

Prepare a direct question with a clear focus. Keep essential subject terms but remove unnecessary wording. Ask follow-up questions separately.

Example — Business

Instead of: “Which pricing strategy would you choose, why would you choose it, and what disadvantages might it have compared with the alternative?”

1. Ask: “Which pricing strategy would you choose?”
2. After hearing the answer: “What is your main reason?”
3. Then: “What is one possible disadvantage?”

This keeps the intellectual challenge while making each part easier to follow.

2. Allow thinking time

After asking a question, pause silently. As a starting point, allow five to ten seconds; give longer when students need to examine information or develop an explanation.

Tell students what the pause is for: "Take a moment to think. I'll invite answers shortly."

Avoid filling the silence with extra questions. If students remain uncertain, check whether they need the question clarified or help with the underlying concept.

Example — Engineering: "Which component is most likely to have failed? Take 30 seconds to examine the diagram and identify one piece of evidence."

3. Give brief preparation before public responses

For a demanding question, give students a short opportunity to prepare before speaking to the whole class.

How to do it

1. Display or state the question.
2. Ask students to note a possible answer and one reason.
3. Give pairs a brief opportunity to compare ideas.
4. Invite contributions to the class.

Keywords are enough; students do not need to write a script.

Example — Public health: "Which intervention would you prioritise in this case? Note your choice and one reason. Then compare with your partner before we hear some responses."

4. Offer simple ways to begin an answer

Display two or three optional sentence openings that match the type of thinking required. Explain what each helps students do and model an answer using course content.

Purpose	Optional opening
Explain a cause	"One possible cause is… because…"
Use evidence	"The data suggest… because…"
Make a qualified claim	"This depends on…"
Compare options	"I would choose… rather than… because…"

Example — Economics: "The data suggest that purchasing power fell because prices increased faster than income."

Students can use their own wording. The openings provide a starting point; they do not determine the answer.

5. Make invitations to answer predictable

Explain how you will invite contributions. Ask the question to everyone, provide thinking time, then select respondents.

For example: “I’ll give everyone time to think, then invite a few people to explain their choice. You can ask for more time if you need it.”

If a student is not ready: “Would you like to hear another response first, and then come back in?”

Return to them as agreed, without treating the delay as a failure. They might contribute a reason, add evidence or comment on the response they heard.

Example — Nursing: “Which action should come first? Everyone, note your choice. I’ll then invite a few people to explain their reasoning.”

6. Help students develop an initial response

Let students finish before responding. If their meaning is unclear, check your interpretation. If the answer is incomplete, ask one focused follow-up.

Example — Environmental science

Student: “Less water… crops.”

Teacher: “Are you saying that reduced rainfall lowers crop yields?”

After the student confirms: “What happens to the plants when water is limited?”

If the subject explanation is inaccurate, identify the point that needs reconsideration:

“You have identified water availability as relevant. Let’s check the effect you described. What does the graph show?”

Acknowledge useful thinking without presenting an incorrect answer as correct.

A short routine to try

Choose one question in your next class:

1. Ask it clearly and display it if possible.
2. Give everyone time to think.
3. Invite a response.
4. Listen without interrupting.
5. Ask one follow-up that helps the student explain their reasoning.

Where rapid responses are part of the learning goal, build towards them through supported practice. Explain when students are practising subject reasoning and when speed of response also matters.

Supporting Resources

Northeastern University, Center for Advancing Teaching and Learning Through Research. (n.d.). *Active question and answer techniques*

University of Minnesota, College of Continuing & Professional Studies. (n.d.). *Teaching support*

University of Waterloo, Centre for Teaching Excellence. (n.d.). *Question strategies*

Washington University in St. Louis, Center for Teaching and Learning. (n.d.). *Asking questions to improve learning*